# CONVERSATIONAL IMAGE RECOGNITION CHATBOT

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Abstract- Conventional chatbots are usually text or speech based and cannot be able to analyze visual data that includes images, diagrams, or complicated real-life scenes. This constraint limits their applicability in any field that requires visual comprehension such as assistive technology, education, and monitoring in industries. The present paper introduces a Conversational Image Recognition Chatbot, which is a multimodal Artificial Intelligence agent, which combines YOLOv8 to detect objects in real-time, BLIP-2 to caption and question images and answer them, and LLaMA-3 to reason about a context. The models are coordinated using a Python based modular structure in which a modular Gradio interface is used and the components used are LangChain/LangGraph to handle multi-turn memory and support natural-language queries in which the user can upload images, ask questions, and obtain both annotated images and written responses. Tests on a variety of real-world images both prove that the object localization is accurate, semantic captions are appropriate, and coherent multi-turn conversational dialogs attached to the visual context. The given system provides the transition between computer vision and natural language interpretation and can be used in smart digital assistants in education, serving as an aid to people with disabilities, in industrial inspection, and human-AI interaction.

Keywords- Multimodal learning, conversational AI, image recognition, visual question answering, vision transformers, large language models.

## INTRODUCTION

The artificial intelligence (AI) has greatly changed the interaction between human beings and computers in the sense that the systems can comprehend human natural language and produce responses that can be likened to their natural speech. Siri, Google Assistant, and Alexa are the examples of conversational agents used in the education, health, online shopping, and support areas. Even with this advancement, the majority of chatbots that have been developed are unimodal and only work with either text or speech. They do not process visual information like pictures, graphs or real-life situations that restricts its usefulness in the process where one needs to think in pictures. Visual and linguistic information are bound together and perceived by humans in describing the world around them. The use of text-based conversational systems limits the scope of activities that can be facilitated in practice. As an illustration, analyzing online shopping product images, comprehension of diagrams in academic materials, object recognition in industrial design, or description of the environment in assistive technology to aid the blind necessitate a strong visual understanding with the language reasoning. Although current computer vision architecture e.g. YOLO and vision-language architecture e.g. BLIP and CLIP have set impressive performance in object detection, recognition, and caption generation, these have not been used broadly in end-to-end conversational agents.

Such a shortcoming brings about the necessity of having multimodal conversational systems signed to have visual perception closely coupled with natural language understanding. This type of systems must be capable of analyzing pictures posted by the user, identifying and locating objects of interest, creating semantic descriptions, responding to visual queries, and participating in a consistent, multi turn conversation based on the visual environment. Most of the research done in the past has treated object detection, image captioning and the visual question answering (VQA) as distinct problems, and a comparative paucity of literature have offered an integrated framework integrating all the three functions into an interactive chatbot. To overcome this issue, this paper will suggest a Conversational Image Recognition Chatbot that will combine three mutually supporting elements (YOLOv8 and quick real-time object detection, BLIP-2, and image captioning and VQA, and LLaMA-3 and context-driven conversational reasoning).

The system is coded in Python and includes Gradio interface, which allows users to feed in a picture, inquire about it using natural language questions, and get annotated picture and a text response in real-time. Outputs (bounding boxes and labels (visual) and captions and dialogue (linguistic)) are imposed together into one experienced multimodal interaction. The principal contributions of the work are as follows: We create a multimodal conversational chatbot based on combining object detection, image captioning, and visual question answering as the same dialogue-based system. We suggest a scalable modular architecture to create YOLOv8 + BLIP-2, and LLaMa-3 together and have scalable deployment and allow them to upgrade independently. To provide intuitive image-based conversations, we apply a real-time interactive interface and use Gradio which provides both annotated images and natural-language explanations. We perform an experimental analysis of a variety of real-life images, including the high level of detection, semantically relevant captions, and logical multi-turn conversations based on the visual context.

## RELATED WORK

The section recounts the previous studies on three topics that are pertinent to the current multimodal conversational system: (A) text-based conversational agents, (B) computer vision and object detection, and (C) vision-language models and multimodal dialogue systems. Conversational Agents based on text. First generation conversational agents ELIZA and ALICE used pattern matching on a rule basis creating inflexible behaviour and a weak ability to generalize outside the templates [1]. As neural sequence-to-sequence (Seq2Seq) model and encoder-decoder architecture were developed, conversational systems became more free and fluent [2]. The development of large language models (LLMs) including GPT, PaLM, and LLaMA also advanced in the field of natural-language understanding and made possible open-domain dialogue, question answering, and reasoning without requiring task-specific fine-tuning [3]. It is based on these developments that modern commercial assistants, such as Siri, Google Assistant, and Alexa, have become. But these systems are very unimodal and can only process text or speech. They are unable to interpret visual information like pictures or videos directly and as such, it is not applicable to activities that involve the interpretation of images like answering questions about pictures posted by users of a site or giving explanations that require visual evidence. This shortcoming encourages the incorporation of visual processing in conversational agents. Computer vision and object detection involves identifying individual objects and employing those to recognize more broad concepts.<|human|>

Object detection and Computer vision Computer vision and object detection refers to recognizing individual objects and using them to represent larger things. Deep learning has allowed the development of computer vision activities like classification, detection, and segmentation, single-Models such as AlexNet, VGG, ResNet, and EfficientNet have performance at the state of the art in tasks like classification [4]. Two wide-known procedures are available to object detection: Such a two-stage detector (e.g., Faster R-CNN) makes region proposals then is classified with high accuracy at higher computational cost [5]. Single-stage detectors (e.g., SSD, YOLO) are directly applied to detecting objects as a real-time regression task, which makes them practical [6]. The YOLO family has been developed quickly, and the YOLOv8 has enhanced the design of the backbone, feature aggregation, and purpose of training to trade-off speed and precision. These sensors find many applications in areas like self-driving and inspection within industries. However, traditional detectors provide structured (bounding boxes, labels, confidence scores) only and cannot address natural-language explanations or can also have a conversational interface. Therefore, in the case of conversational uses, object detection should be accompanied by a language model that is able to reason and speak.

Vision-Language Models, Multimodal Conversational Systems. Vision-language models are supposed to learn to evaluate both visual and textual information. Initial captioning was based on CNN encoders and RNN decoders to produce image descriptions [7]. Subsequently, mechanisms of attention and transformer constructions enhanced visual-text alignment, which allowed captions with more context. Other cross-modal models like CLIP also developed cross-modal learning with the goal of training on large image-text datasets to produce common embedding spaces, which enabled zero-shot classification and retrieval [8]. BLIP and BLIP-2 builds on this body of work: they use vision encoders with lightweight query transformer modules to communicate with frozen LLMs, which facilitates effective multimodal reasoning on prompts like captioning and VQA [9]. Parallel to this, there are now multimodal conversational systems that combine vision models with LLMs that enable users to ask questions about images and receive answers in natural language. Nevertheless, most of the available systems have the following major limitations: Their use is more of research prototypes and not real-time, practical tools that involve detection, captioning and dialogue in a single interface. Visual processing is traditionally considered a one-step object-free feature extraction task, where object-level detections are never made explicit and can benefit an object-level grounded reasoning. Conversational memory is small and therefore it minimizes the context of multiple queries on the same image. These deficiencies motivate the necessity to have the system that integrates real-time object detection, caption generation, VQA, and multi-turn conversational reasoning in one multimodal system- a goal that our proposed system fulfills.

## METHODOLOGY

The suggested Conversational Image Recognition Chatbot is publicly powered by the modular multimodal architecture that integrates object recognition, vision-language perception, and dialogue based on large language models. The system severs the pre-trained YOLOv8 models, BLIP-2 models, LLaMA-3 models in a Python approach with Gradio interface.

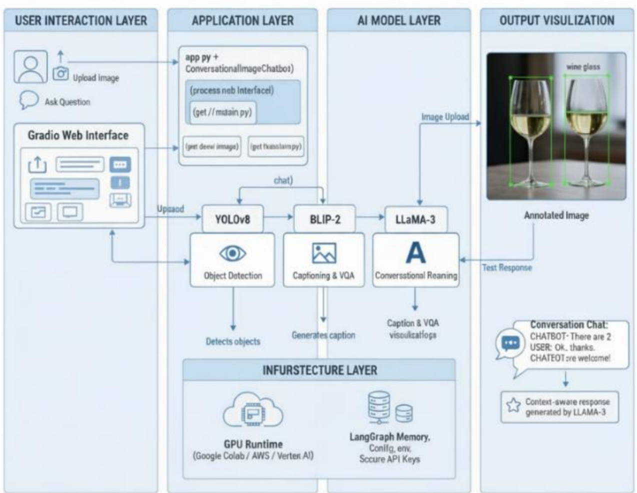


Fig 1. System Architecture

Overall Architecture There are four components of the system: User Interface: An upload image, question entry, and an image annotation and chat history web interface built using Gradio. Vision Layer: The image uploaded is processed by YOLOv8 (real-time object detection) YOLOv8 in terms of broing boxes (bounding boxes), real-time labels and real-time confidence score determine object positions in a frame. Image captioning (BLIP-2) as well as, when necessary, visual question answering (VQA). Language & Reasoning Layer: The final response provided by LLaMA-3 is based on a prompt which contains YOLOv8 detections, BLIP-2 captions/VQA results, the query at hand, and the history of conversation. Controller & Memory: A Python controller is to direct inputs, construct multi-modal prompts and store vision responses as well as multi-turn dialogue history.

Vision Processing The processing of images as tensors into OpenCV/NumPy/PyTorch uses open technologies to diminish their size, normalize them, and transform them. YOLOv8 Detection: YOLOv8 is pre-trained, which presents visual information in structure. Detections can also be utilized after non-maximum suppression to draw annotated images as well as to generate an adapted text summary (e.g., 2 people, 1 car observed). BLIP-2 Captioning & VQA: Scene captions of new images are created by BLIP-2. In the case of a visually-grounded question (e.g. How many cars), BLIP-2 provides a short VQA response. These generate the semantics context in the grounding of a conversation.

Conversational Reasoning During every truncheon the controller will create a multimodal prompt with: YOLOv8 object summary, BLIP-2 caption or VQA answer, The user's question, and Relevant history of conversation. A grounded response is produced by LLaMA-3 which is advised not to produce hallucinations and be consistent with previous context. The new turn is buffered to more than one turn. Pros of the Proposed Design. Better than the chatbots that rely on text only: The responses are not based on guesses but rather on YOLOv8 and BLIP-2 results. Interactive as opposed to standalone vision models: The explanation, follow-up questions, and natural conversation are added by LLaMA-3. Flexible compared to Monolithic multimodal: Updating each of the modules (YOLOv8, BLIP-2, LLaMA-3) can be done separately. Practical and real-time: Fast execution with small training resources is guaranteed using pre-trained models.

## SYSTEM OVERVIEW

This section gives a summary review of the proposed Conversational Image Recognition Chatbot. It describes the problem under discussion first and then gives a description of the entire workflow that is undergone by the system starting with user input to system output.

Problem Definition: The fundamental purpose of the system will be to allow a user to engage in a natural conversation regarding uploaded photograph. The user should be able to: Upload any picture, ask questions of what you can see in the picture, and Receive: A label and bounding box image of the objects detected. An answer written in a natural language, matching what is contained in image and fits the context of the conversation taking place. To help with the conversational understanding: The visual input is the picture uploaded. The current query is the question that is asked by the user in this turn. The conversation history is a record of all the past questions and answers so that the system is the same throughout the various turns. The system produces: A list of the object detections of YOLOv8 (bounding boxes, labels, and the confidences scores). A text reply produced by a massive language model, relying on the pictures on the images, the question posed by the user, and the conversation that is taking place. In general, the chatbot acts like a function which accepts: an image, the user's question, and the previous conversation, and returns: detected objects and a relevant textual answer. YOLOv8 offers when an object is identified, BLIP-2 offers captions and visual comprehension whereas LLaMA-3 generates the eventual conversational reply.

System Workflow: The system is non-linear with a pipeline that makes use of computer vision, vision-language processing, and conversational reasoning. The stages involved in the workflow are: The interface and input collection; the interface consists of the front-end of the website and forms for user input.

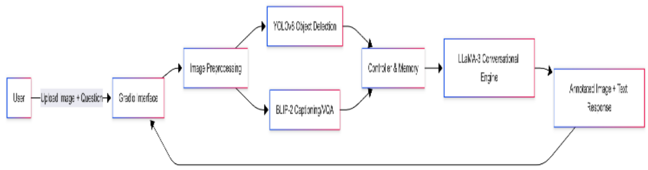


Fig 2. System Workflow

User Interface and Input collection: The user interface is the front of the site and/or input forms to capture user input (Grogan 78). To communicate with the user, there is an interface in the form of a Gradio web interface. It allows users to: Upload an image, There are questions, type and Send the entire history of the conversation.

Visual Image Process and Extraction of Features. On the uploading of a new image the system: Normalizes and downsizes the image in advance. Sends it to: YOLOv8 that identifies objects and provides bounding boxes, labels and confidence scores. BLIP-2 that produces the initial caption of the scene. By this time, the system has both: Visual information (detections) structured, Semiotic visual data (caption and features).

Visual Question Answering (to be provided only in necessary occasions) In case the question posed by the user is clearly visual content. (e.g. how many people are there, what is the color of the car), BLIP-2 is used in VQA mode. It can generate a brief response that will enable the system to know what the user is querying it with. This response is inferred to provide additional context to the process of creating an accurate final response.

Constructing the Multimodal Context. All the information that is available is combined in a controller module: YOLOv8 detections BLIP-2 caption Optional VQA answer Current user question Conversation history It is all transformed into a single prompt that is well structured and describes: What do we see in the picture, What the user is asking, and What has been already explained. This makes the language model have full and proper context.

Inventory of the Conversational Response. The resulting constructed prompt is forwarded to LLaMA-3 that generates the final natural-language response. The model is instructed to: Found its reactions just on the picture, never speak of a non-existent object, and Will adhere to past conversationals.

Fixing the User Interface and Repository History. Having generated the response: An annotated image is created by drawing bounding boxes and labels of the original image in the position of the YOLOv8. In Gradio interface, there is: The updated annotated image The generated text response The pair (question and reply) is signified with history of conversation, which makes it continuous.

Multi-Turn Interaction The user can proceed to ask questions without the need of re- uploading the picture. The system: Reuses prior findings of detection and caption data, Makes new prompts based on the expanding history of conversation, and Follow-up questions include responses to questions like. Helic- What color of the car you say you had? or "Is that something behind the man? This enables easy, multi-rotation, text dialogue that is image based.

## RESULTS AND DISCUSSION

In this part, the researcher explains the performance of the recommended Conversational Image Recognition Chatbot according to the qualitative test with various images in the real world. The assessment is on object detection, captioning, quality of VQA, conversational coherence, and responsiveness of the system.

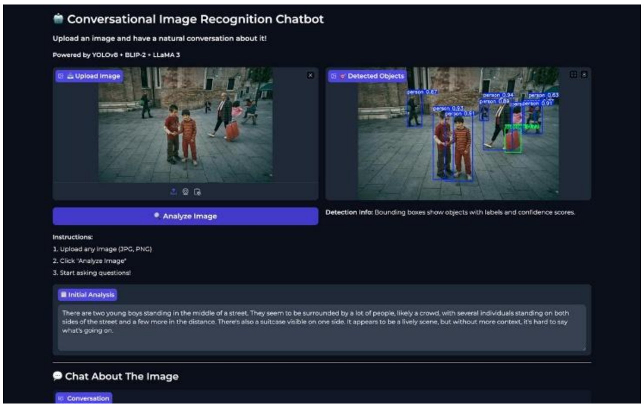


Fig 3. Example - People image

Qualitative Results: The system was able to rightly detect and create useful visual descriptions for objects, interactions between people and objects, as well as the outdoor scenes. Object Detection: YOLOv8 detector was always able to identify key items, including people, vehicles, and ordinary objects. In medium-object images, the annotations were easy to read and discern. Scene Captioning: Captions produced by BLIP-2 made coherent summaries of the picture, and generally included around the subjects of the image as well as inside of the image (e.g., a person near a car, a desk with a laptop and books). Visual Question Answering: In direct visual queries (counts, colors, simple actions), VQA answers of BLIP-2 were generally accurate (identifying something like the number of people or whether an individual was sitting or standing). On the combination of these visual cues with the controller and transferred to the LLM, the chatbot generated well-grounded and fluent responses with references to specific objects, instead of generating generic textual responses. An example of detection and captioning performance on a real-world outdoor scene is shown in Figure 3.

Conversational Quality: Multi-turn interaction exhibited a good conversational behaviour: Relevance: The system normally provided an answer to the question of the user with both detections and captions used to explain activities or things on the scene. Grounding: Since the prompt included clear identifications and BLIP-2-generated responses, LLaMA-3 generated responses that were closely related to the visible content, and a small percentage of visuals were mentioning non-existent objects. Multi-turn Coherence: The chatbot responded to follow-up questions (e.g. what color the aforementioned car is, is a person sitting/standing, etc.), which demonstrates that it has a productive short-term conversation memory. These findings suggest that the integration of YOLOv8, BLIP-2, and LLaMA-3 with explicit history control can be used to help maintain consistent image-grounded dialogue.

Practical Responsiveness: The system proved to be interactive in response time operations befitting usability applications. Only in case of the upload of a new image, heavy computation (YOLOv8 and BLIP-2) was performed. The later turns were based on already cached detections and features, instead asking a language-model call. This maintained low delays and also made the interface responsive during a conversation. Making it work and what benefits have failed: Although the overall performance was being good, a number of limitations were identified: Senoc Fly by first or Secondary detections: YOLOv8 would occasionally miss small or cluttered objects, which would result in partial or poor description of conversations. Caption/VQA Ambiguities: Also in complicated scenarios, BLIP-2 tended to pay attention to little things or give imprecise VQA responses; the ambiguities sometimes transferred to the response of the chatbot. Poor Evidence of High-level Reasoning: Although LLaMA-3 was capable of doing simple reasoning, it was unable to do abstract reasoning or heavily-context based questions other than what could be seen. Domain Dependence: Because every model is trained, they got worse on task-specific images (impediments such as medical scans or technological schemes) which were not like training cases. These shortcomings point to areas of improvement (e.g. domain-specific fine-tuning and incorporating other tools like OCR or segmentation).

Evaluation as Compared to other methods:

Versus Text-only Chatbots: LLAMs are text-based and are incapable of seeing and hallucinatory in nature. The proposed system does not do this by basing the responses on explicit detections and the BLIP-2 outputs. Opposite Standalone Vision Models: Vision models are capable of contemplating images but capturing them or labeling them but unable to talk or respond to questions. Explanatory responses and interactive responses can be added with the addition of LLaMA-3. Opposing Monomodel Multimodel models: End-to-end multimodal models are good, yet difficult to manage or modify. To improve each of the components (YOLOv8, BLIP-2, LLaMA-3) we have a modular design that enables each to be tested and refined individually. On the whole, findings indicate the usefulness of the modular architecture to offer a practical, interpretable and effective image-grounded conversational AI solution.

## CONCLUSION AND FUTURE WORK

This paper introduced a Conversational Image Recognition Chatbot which combined the current vision and language models in a modular multimodal pipeline. Using periodic Gradio input/output, the system brings real-time detection (YOLOv8), captioning, and VQA (BLIP-2) and context-conditional dialogue (LLaMA-3) together in one system. In contrast to text-based assistants or even isolated vision models, it can read images provided by users, respond to image-based questions and allow multiturn image-grounded conversations. In diverse real-world images, qualitative assessment indicated that YOLOv8 was consistent in identifying large objects, BLIP-2 generated meaningful captions and VQA responses and LLaMA3-generated coherent responses that were grounded to both image and history of the conversation. The modular architecture also provided the system with an easier analysis, maintenance and extension. Although the system has strong points, it also shows certain weaknesses among them being the missed captures in crowded scenes, vague captions/VQA on complicated pictures, as well as worse results with specialized data than with the pre-training datasets. The following are possible directions in the future based on these concerns: Domain-specific adaptation: Fine-tuning of the vision and vision-language models towards specific domains: e.g. medical or industrial images. Multimodal capability at scale: Add support of richer and structured visual reasoning with the expansion of OCR, segmentation or depth estimation. increase conversational memory: Enhancement of long-term context management, cross-image referencing and bringing in external, factual information. User studies and quantitative analysis:hspace/ Perform wider testings in order to gauge user satisfaction, accuracy and comparison with other multimodal systems. On the whole, this findings indicate that a thoughtful incorporation of the latest vision and language models can foster a useful and feasible system of image-guided conversational AI, and potentially it may be applied in education, accessibility and interactive systems in the real world.

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